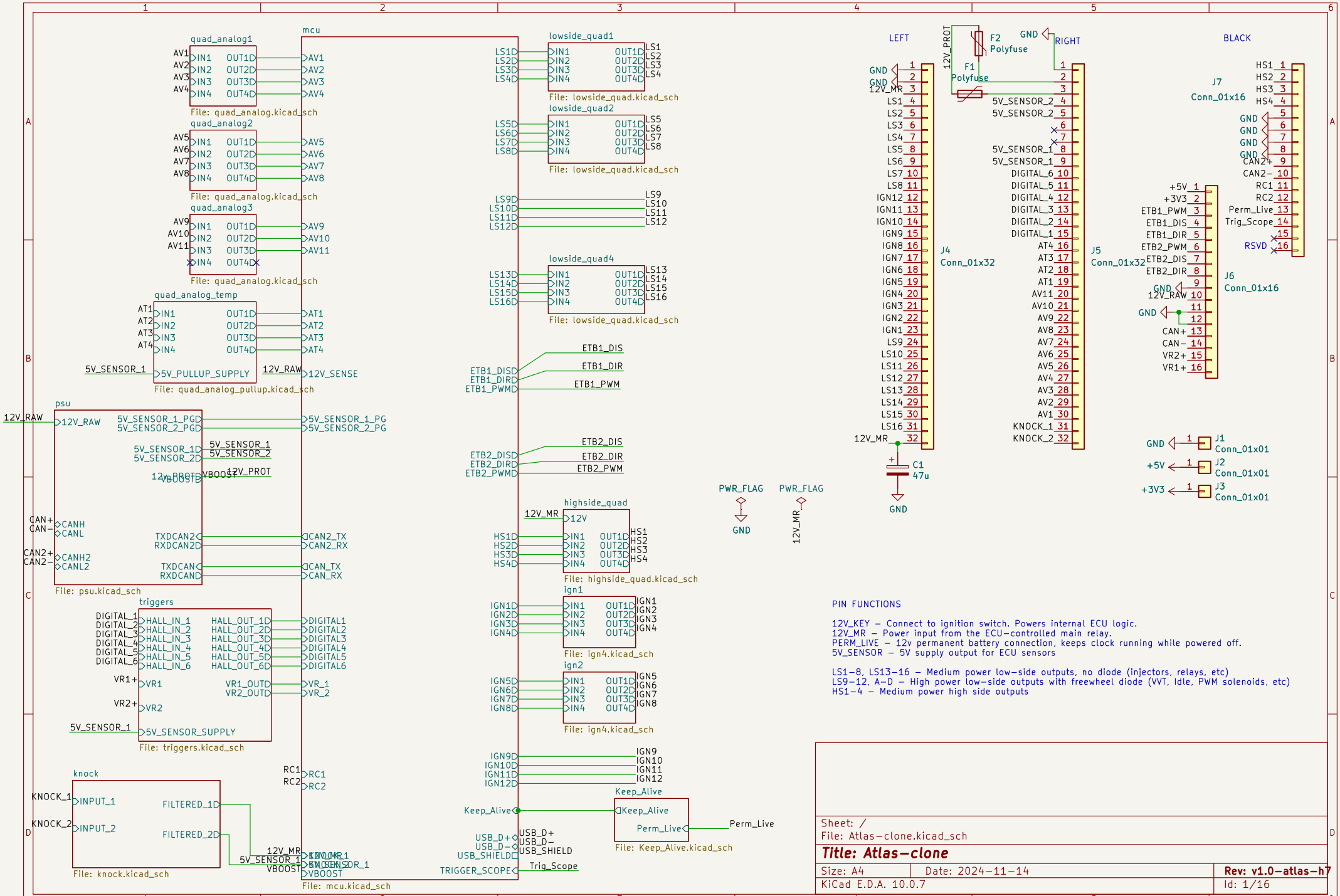
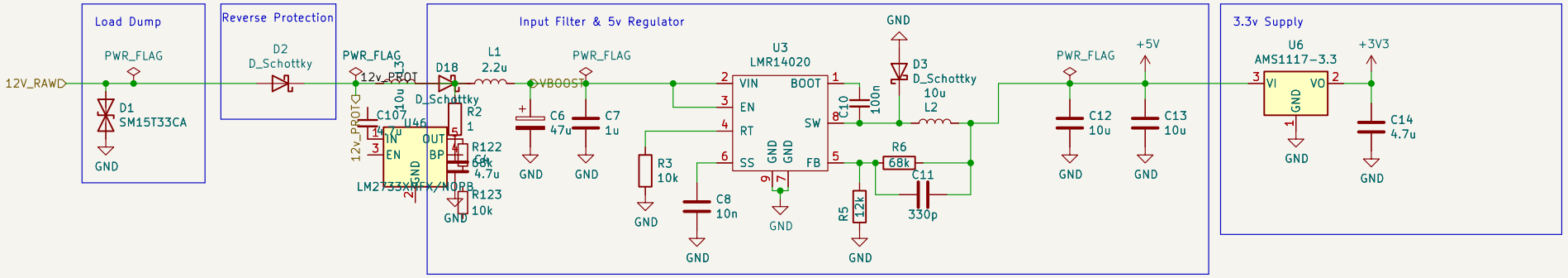
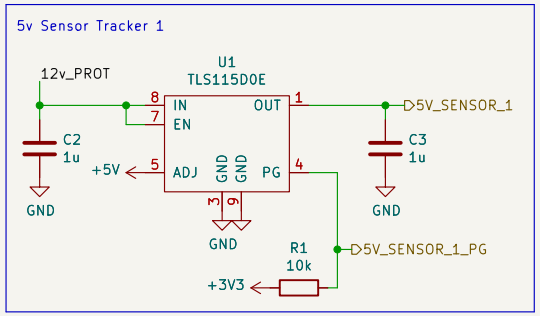
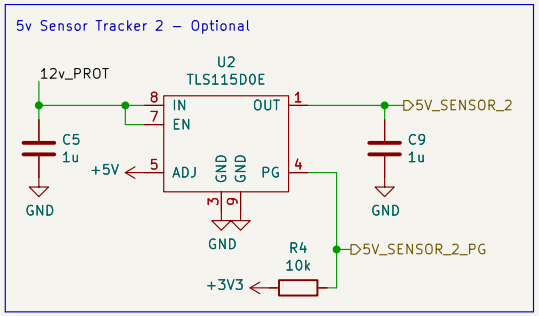
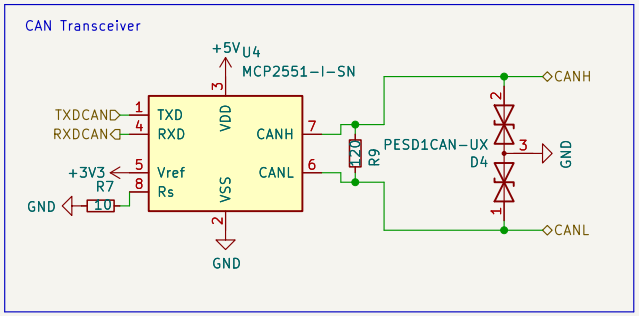
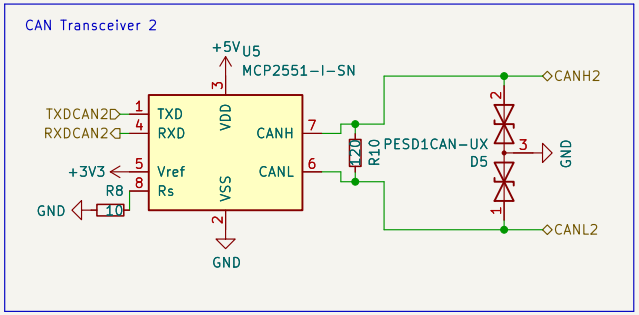

quad_analog1
File: quad_analog.kicad_sch
quad_analog2
File: quad_analog.kicad_sch
quad_analog3
File: quad_analog.kicad_sch
quad_analog_temp
File: quad_analog_pullup.kicad_sch
mcu
File: mcu.kicad_sch
lowside_quad1
File: lowside_quad.kicad_sch
lowside_quad2
File: lowside_quad.kicad_sch
lowside_quad4
File: lowside_quad.kicad_sch
highside_quad
File: highside_quad.kicad_sch
ign1
File: ign4.kicad_sch
ign2
File: ign4.kicad_sch
Keep_Alive
File: Keep_Alive.kicad_sch
psu
File: psu.kicad_sch
triggers
File: triggers.kicad_sch
knock
File: knock.kicad_sch
LEFT
RIGHT
BLACK
J4
Conn_01x32
J5
Conn_01x32
J6
Conn_01x16
J7
Conn_01x16
F1
Polyfuse
F2
Polyfuse
C1
47u
PWR_FLAG
PIN FUNCTIONS
12V_KEY – Connect to ignition switch. Powers internal ECU logic.
12V_MR – Power input from the ECU-controlled main relay.
PERM_LIVE – 12v permanent battery connection, keeps clock running while powered off.
5V_SENSOR – 5V supply output for ECU sensors
LS1-8, LS13-16 – Medium power low-side outputs, no diode (injectors, relays, etc)
LS9-12, A-D – High power low-side outputs with freewheel diode (VVT, Idle, PWM solenoids, etc)
HS1-4 – Medium power high side outputs
Sheet: /
File: Atlas-clone.kicad_sch
Title: Atlas-clone
Size: A4
Date: 2024-11-14
Rev: v1.0-atlas-h7
KiCad E.D.A. 10.0.7
Id: 1/16









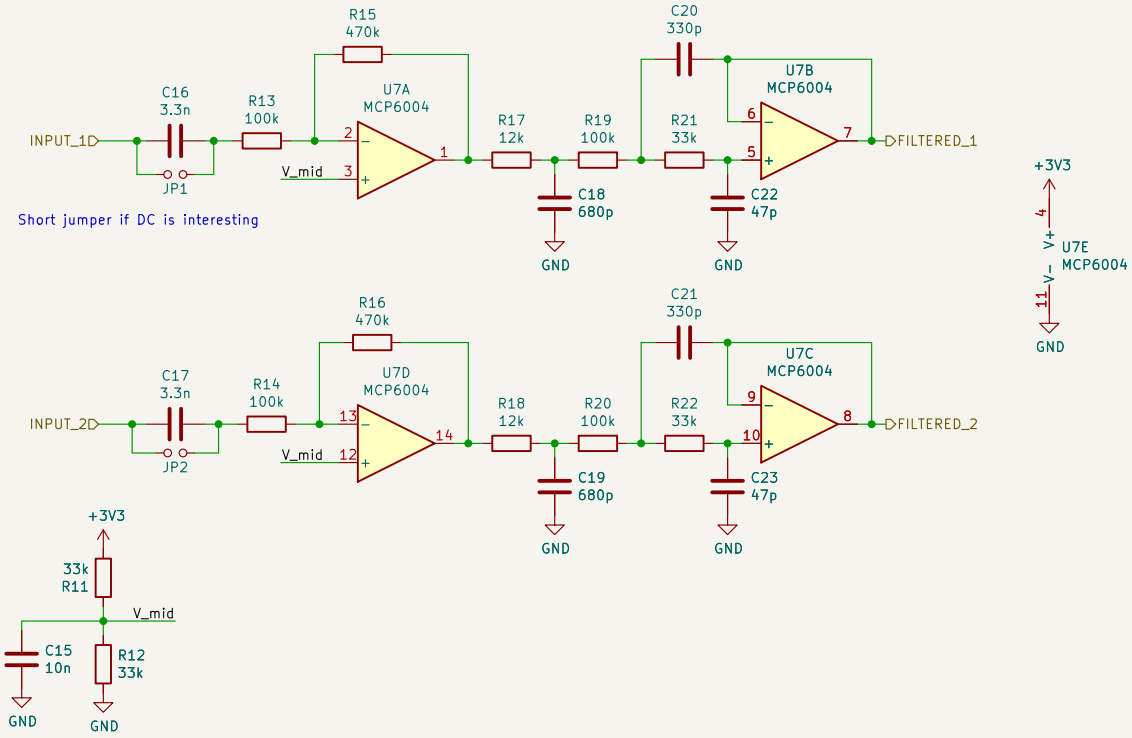


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Sheet: /psu/
File: psu.kicad_sch

Title: Atlas-clone

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 2/16

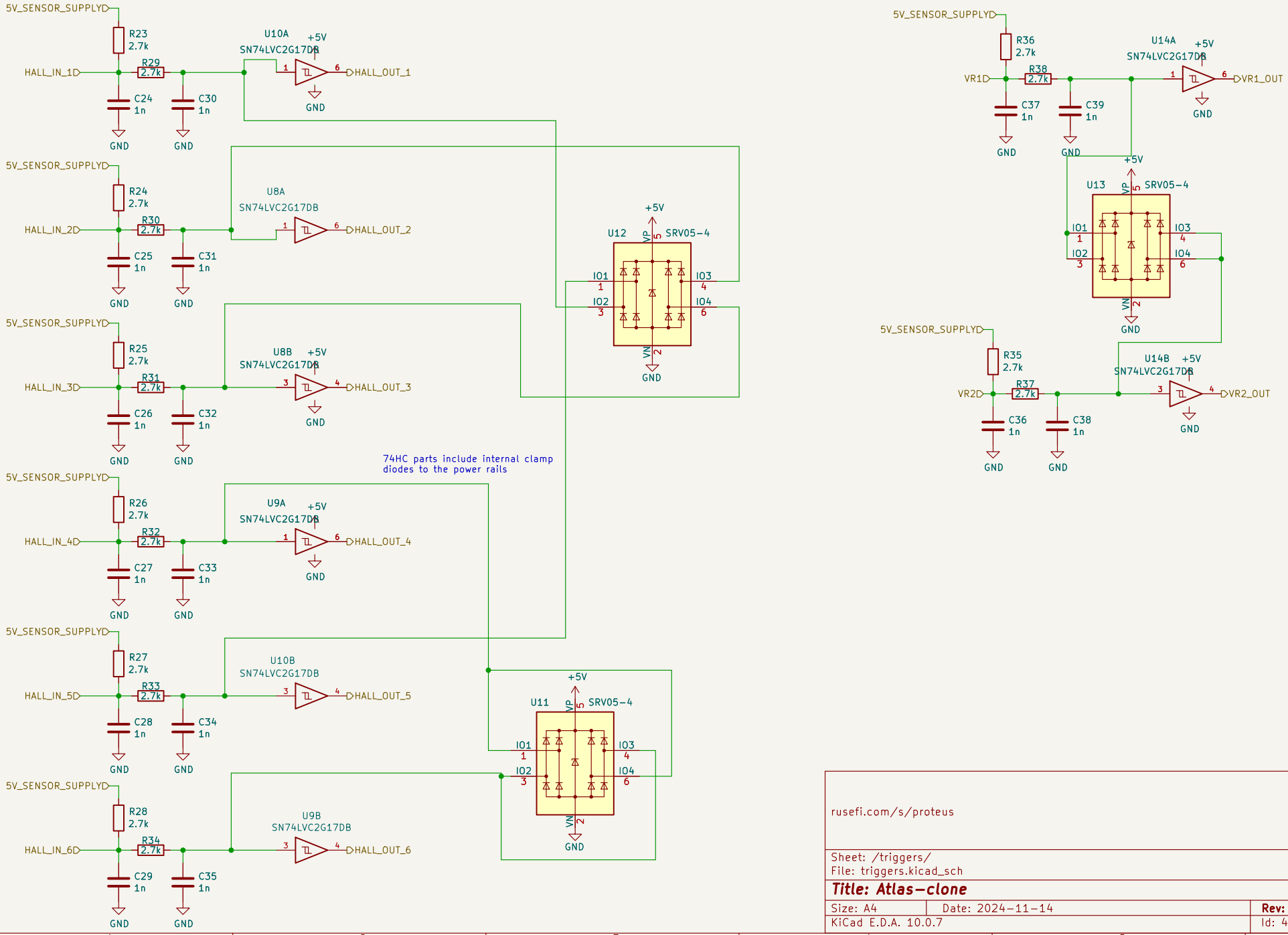


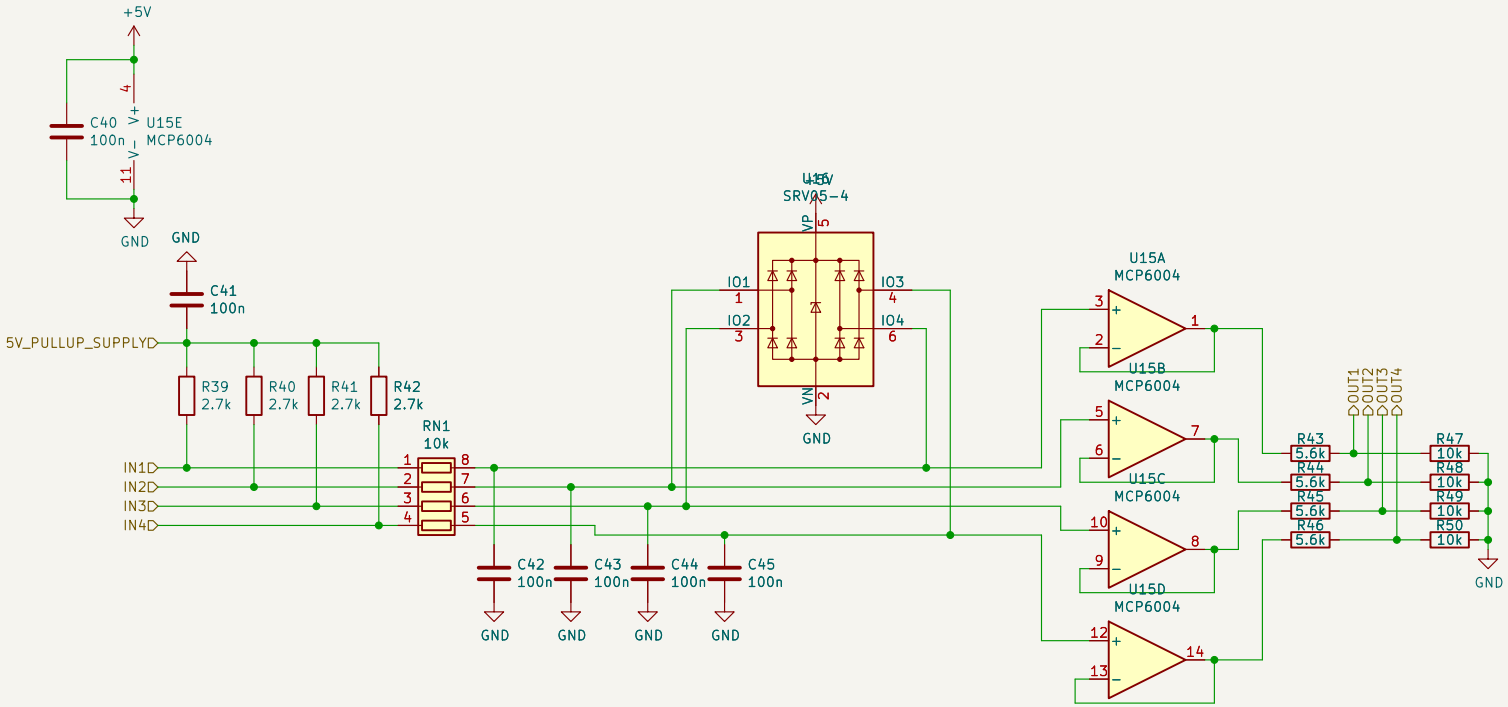
rusefi.com/s/proteus

Sheet: /knock/
File: knock.kicad_sch

Title: Atlas-clone

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 3/16


5V_SENSOR_SUPPLY
HALL_IN_1
HALL_IN_2
HALL_IN_3
HALL_IN_4
HALL_IN_5
HALL_IN_6
R23 2.7k
R24 2.7k
R25 2.7k
R26 2.7k
R27 2.7k
R28 2.7k
R29 2.7k
R30 2.7k
R31 2.7k
R32 2.7k
R33 2.7k
R34 2.7k
C24 1n
C25 1n
C26 1n
C27 1n
C28 1n
C29 1n
C30 1n
C31 1n
C32 1n
C33 1n
C34 1n
C35 1n
GND
U10A
U8A
U8B
U9A
U10B
U9B
SN74LVC2G17DB
+5V
HALL_OUT_1
HALL_OUT_2
HALL_OUT_3
HALL_OUT_4
HALL_OUT_5
HALL_OUT_6
U12
U11
SRV05-4
VP
VN
IO1
IO2
IO3
IO4
74HC parts include internal clamp diodes to the power rails
R36 2.7k
R35 2.7k
R38 2.7k
R37 2.7k
C37 1n
C36 1n
C39 1n
C38 1n
VR1
VR2
U14A
U14B
U13
VR1_OUT
VR2_OUT
rusefi.com/s/proteus
Sheet: /triggers/
File: triggers.kicad_sch
Title: Atlas-clone
Size: A4
Date: 2024-11-14
Rev: v1.0-atlas-h7
KiCad E.D.A. 10.0.7
Id: 4/16

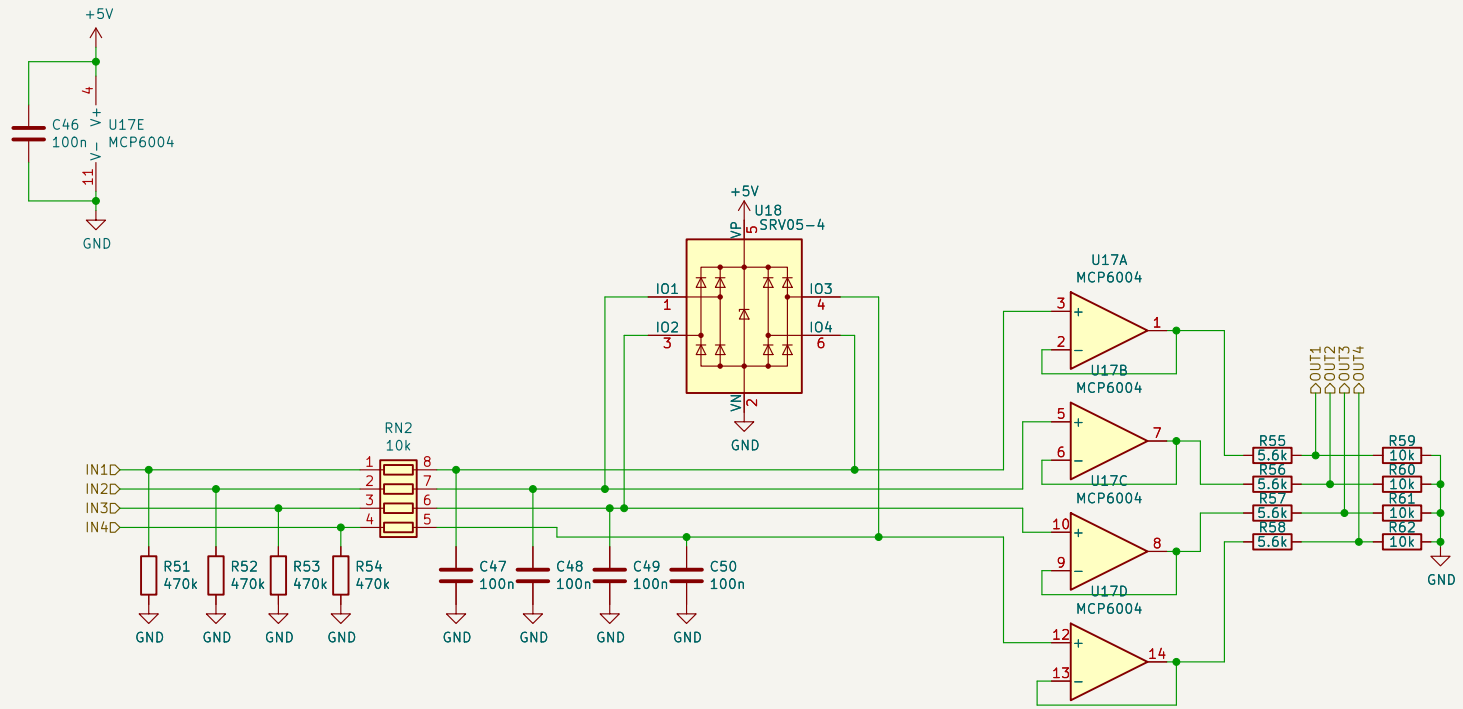


rusefi.com/s/proteus

Sheet: /quad_analog_temp/
File: quad_analog_pullup.kicad_sch

Title: Atlas-clone

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 5/16

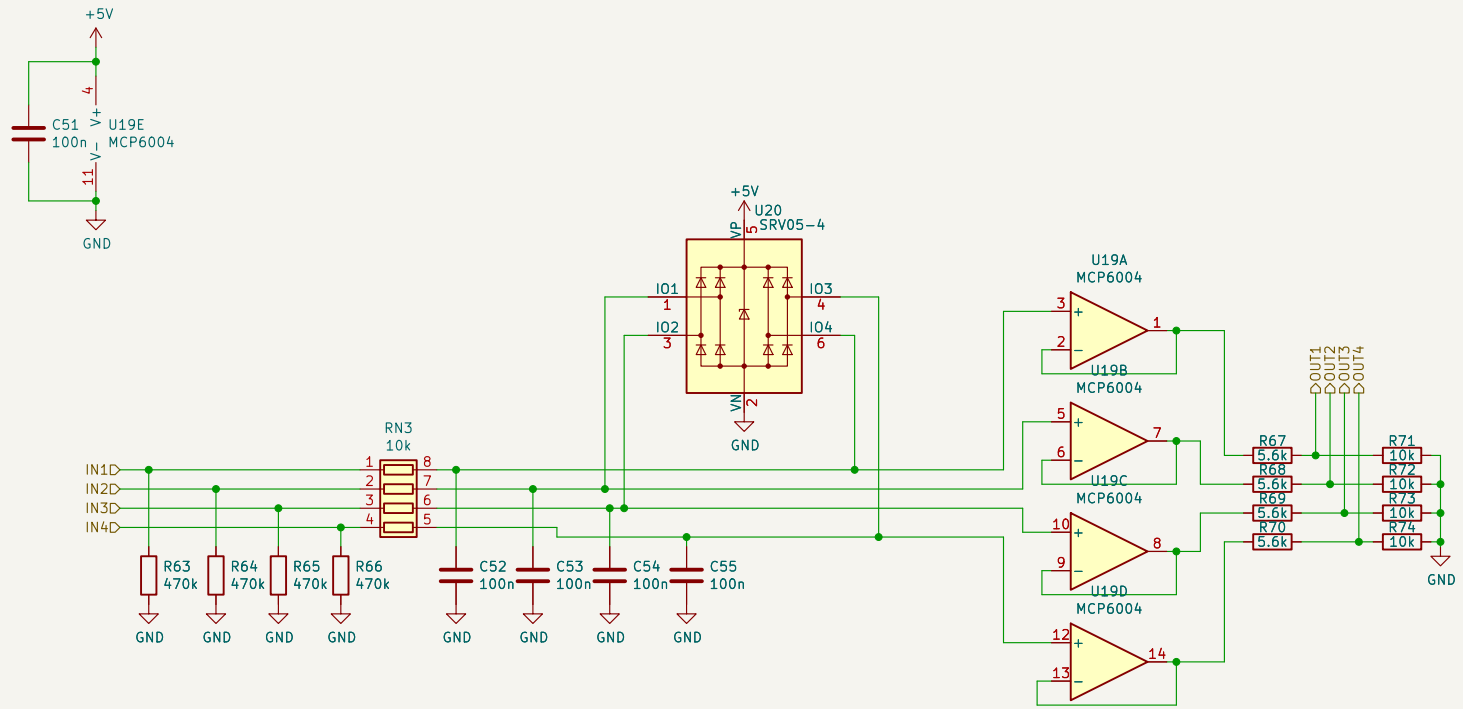


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Sheet: /quad_analog1/
File: quad_analog.kicad_sch

Title: Atlas-clone

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 6/16

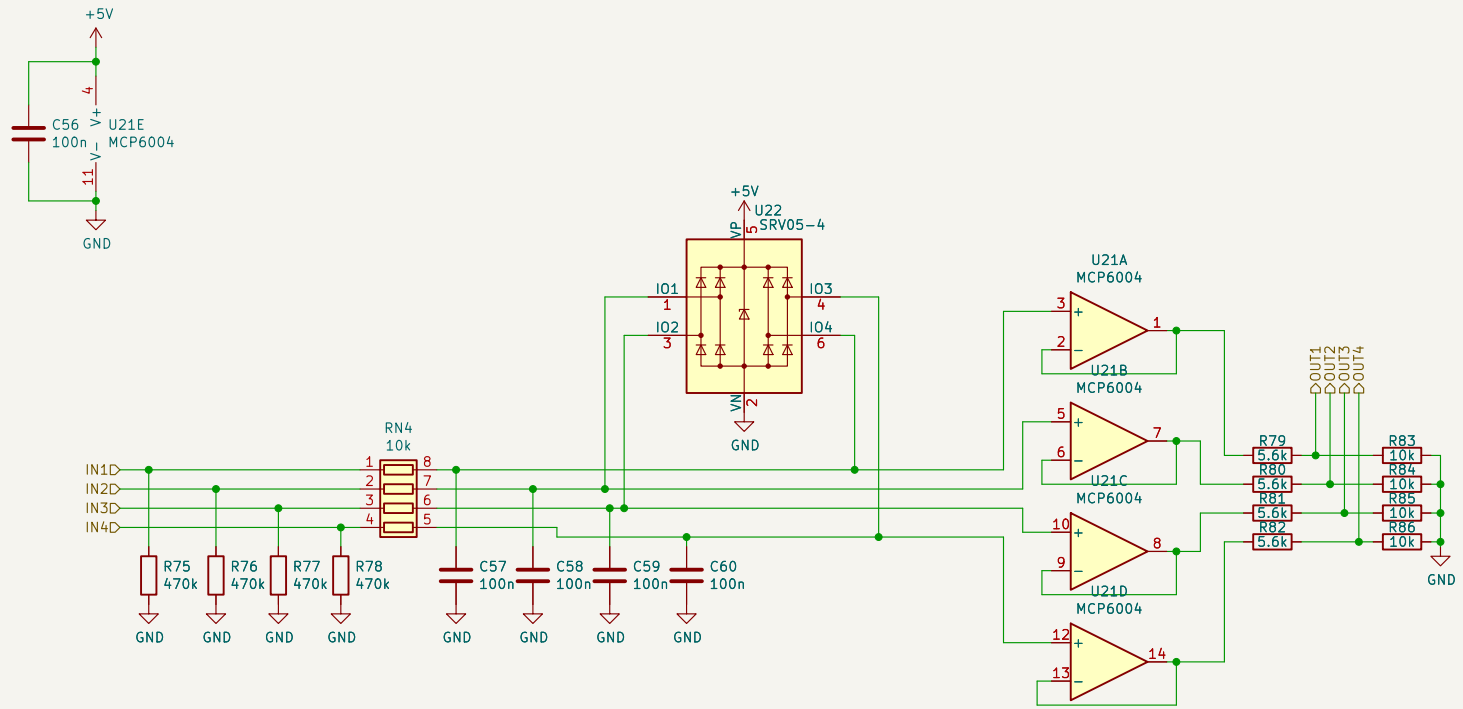


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Sheet: /quad_analog2/
File: quad_analog.kicad_sch

Title: Atlas-clone

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 7/16

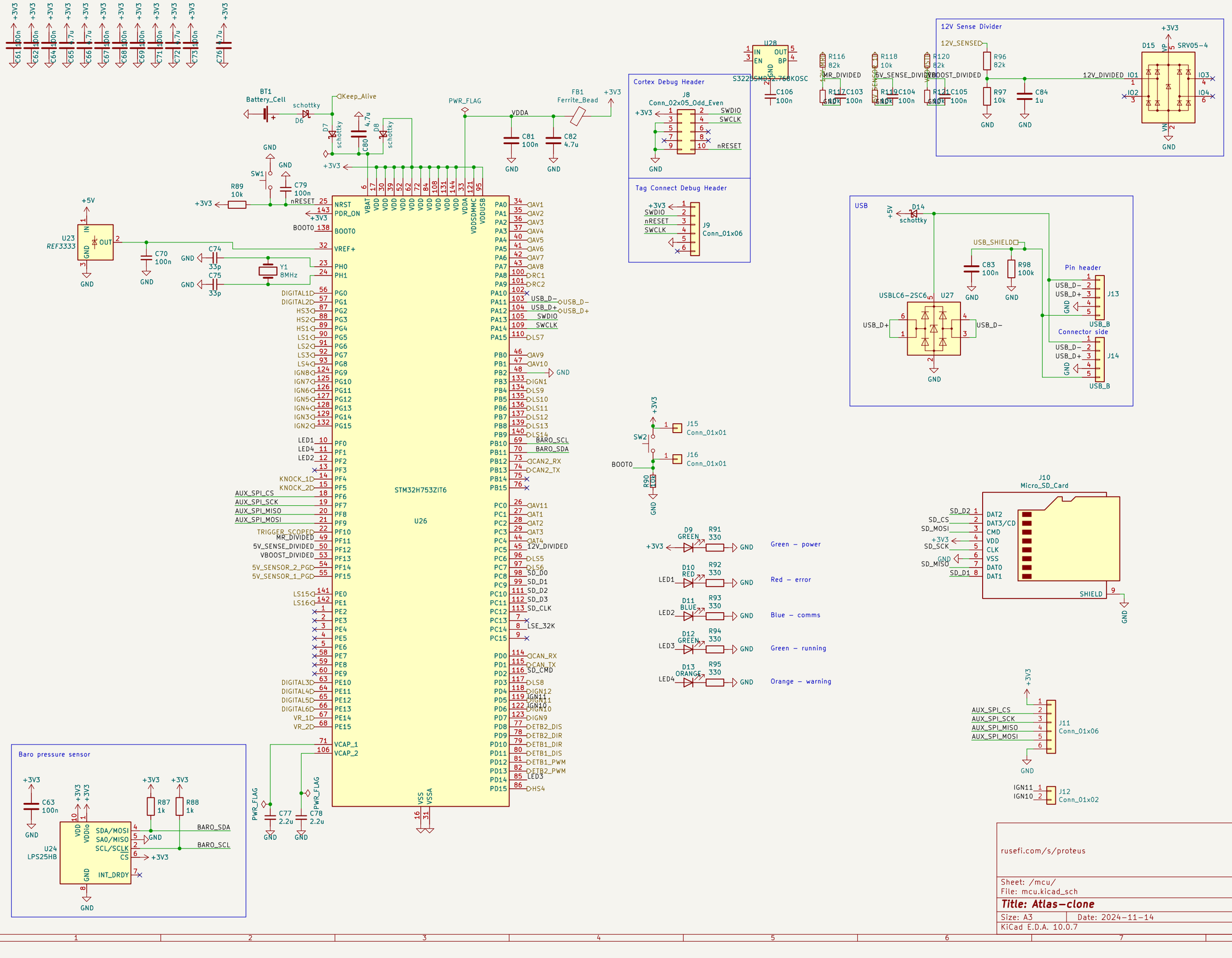


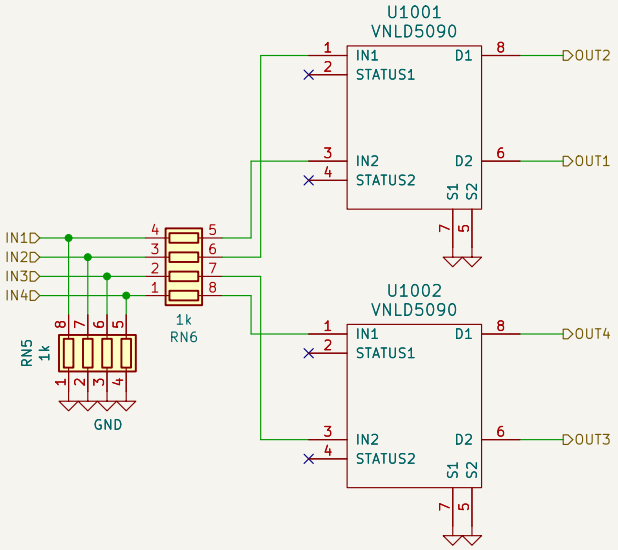
rusefi.com/s/proteus

Sheet: /quad_analog3/
File: quad_analog.kicad_sch

Title: Atlas-clone

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 8/16


C61 100n
C62 100n
C64 100n
C65 4.7u
C66 4.7u
C67 100n
C68 100n
C69 100n
C71 100n
C72 4.7u
C73 100n
C76 4.7u
BT1
Battery_Cell
D6
schottky
Keep_Alive
D7
D8
C80 4.7u
PWR_FLAG
VDDA
FB1
Ferrite_Bead
C81 100n
C82 4.7u
SW1
C79 100n
R89 10k
nRESET
BOOT0
U23
REF3333
C70 100n
C74 33p
C75 33p
Y1 8MHz
STM32H753ZIT6
U26
NRST
PDR_ON
BOOT0
VREF+
PH0
PH1
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14
PC15
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PD8
PD9
PD10
PD11
PD12
PD13
PD14
PD15
PE0
PE1
PE2
PE3
PE4
PE5
PE6
PE7
PE8
PE9
PE10
PE11
PE12
PE13
PE14
PE15
PF0
PF1
PF2
PF3
PF4
PF5
PF6
PF7
PF8
PF9
PF10
PF11
PF12
PF13
PF14
PF15
VCAP_1
VCAP_2
U28
S3225SM032.768KOSC
C106 100n
R116 82k
R117 100n
R118 10k
R119 100n
R120 82k
R121 100n
C103 100n
C104 100n
C105 100n
12V Sense Divider
12V_SENSED
R96 82k
R97 10k
C84 1u
D15
SRV05-4
12V_DIVIDED
Cortex Debug Header
J8
Conn_02x05_Odd_Even
SWDIO
SWCLK
nRESET
Tag Connect Debug Header
J9
Conn_01x06
USB
D14
schottky
USB_SHIELD
C83 100n
R98 100k
USBLC6-2SC6
U27
Pin header
J13
J14
USB_B
Connector side
J15
Conn_01x01
J16
Conn_01x01
SW2
R90 10k
D9 GREEN R91 330
D10 RED R92 330
D11 BLUE R93 330
D12 GREEN R94 330
D13 ORANGE R95 330
Green – power
Red – error
Blue – comms
Green – running
Orange – warning
J10
Micro_SD_Card
SD_D2
SD_CS
SD_MOSI
SD_SCK
SD_MISO
SD_D1
SHIELD
J11
Conn_01x06
AUX_SPI_CS
AUX_SPI_SCK
AUX_SPI_MISO
AUX_SPI_MOSI
J12
Conn_01x02
IGN11
IGN10
Baro pressure sensor
U24
LPS25HB
C63 100n
R87 1k
R88 1k
BARO_SDA
BARO_SCL
C77 2.2u
C78 2.2u
rusefi.com/s/proteus
Sheet: /mcu/
File: mcu.kicad_sch
Title: Atlas–clone
Size: A3
Date: 2024–11–14
Rev: v1.0–atlas–h7
KiCad E.D.A. 10.0.7
Id: 9/16



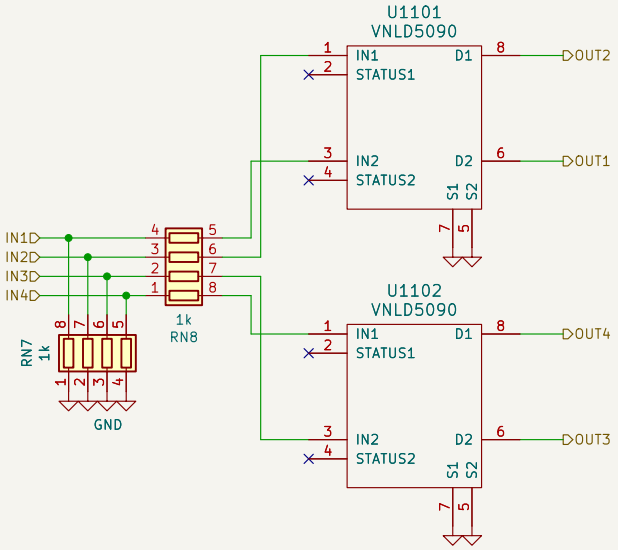
Locate output caps near connector pin

rusefi.com/s/proteus

Sheet: /lowside_quad1/
File: lowside_quad.kicad_sch

Title: Atlas-clone

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 10/16



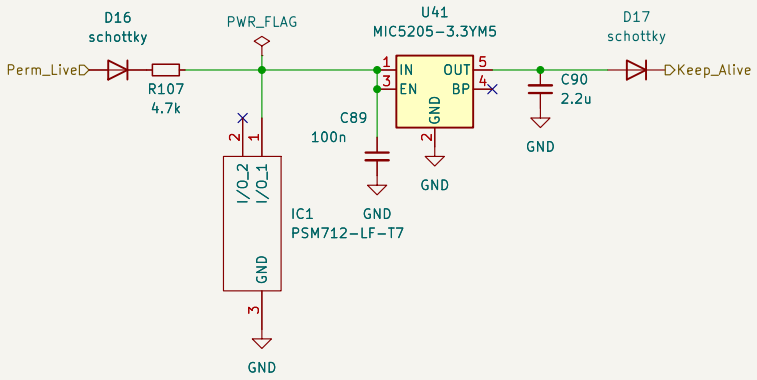
Locate output caps near connector pin

rusefi.com/s/proteus

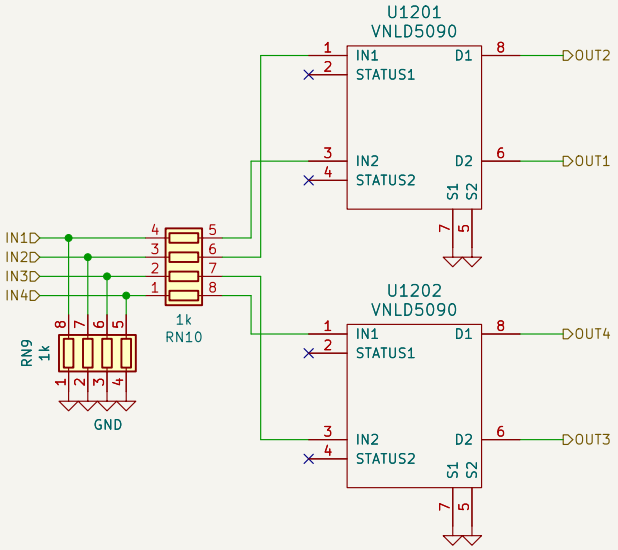
Sheet: /lowside_quad2/
File: lowside_quad.kicad_sch

Title: Atlas-clone

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 11/16



Sheet: /Keep_Alive/		
File: Keep_Alive.kicad_sch		
Title: Atlas-clone		
Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 12/16



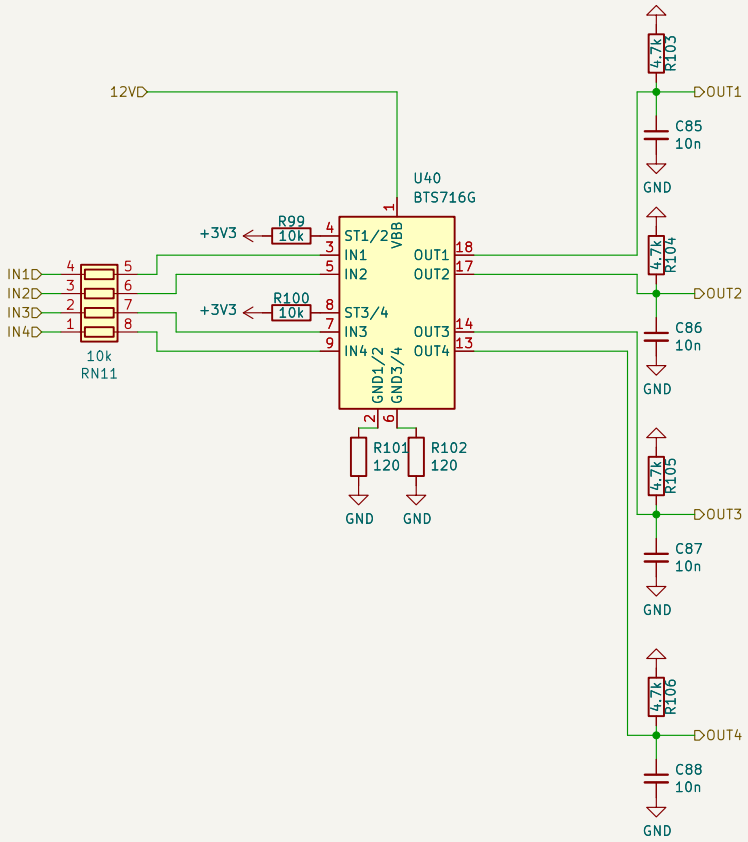
Locate output caps near connector pin

rusefi.com/s/proteus

Sheet: /lowside_quad4/
File: lowside_quad.kicad_sch

Title: Atlas-clone

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 13/16

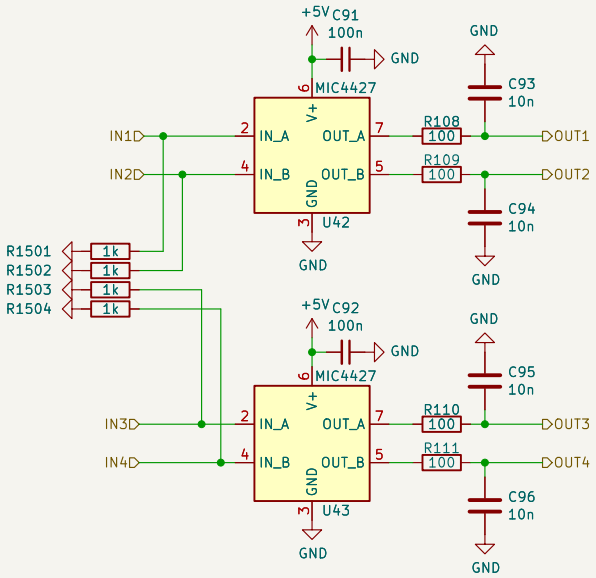


rusefi.com/s/proteus

Sheet: /highside_quad/
File: highside_quad.kicad_sch

Title: Atlas-clone

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 16/16

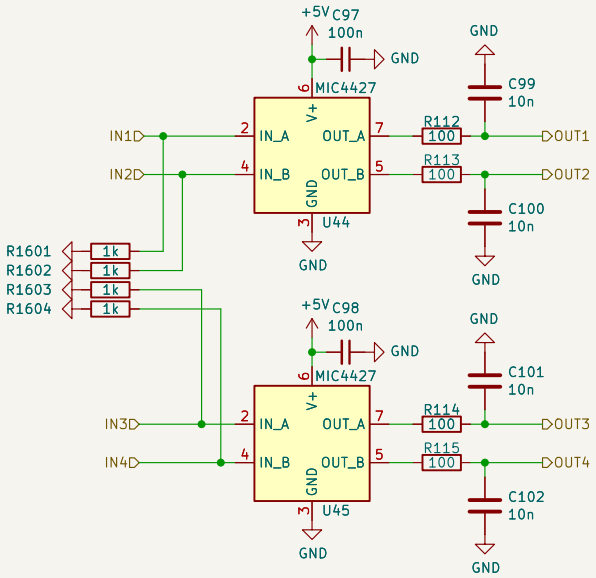


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Sheet: /ign1/
File: ign4.kicad_sch

Title: Atlas-clone

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 17/16



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Sheet: /ign2/
File: ign4.kicad_sch

Title: Atlas-clone

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 18/16